Enhancing Driving Risk Prediction in UBI Systems Through Location and Weather Data Integration

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***Abstract*—The goal of this project is to improve the driving risk prediction by integrating the meteorological and environmental data with telematics data. The research uses many kinds of the information and such as weather, road conditions, and vehicle parameters to create the machine learning models that accurately predict driving risk. To ensure the high-quality inputs for the models and extensive data preparation was done to clean, standardize, and engineer features. To maximize the performance and hyperparameter improving was used for Random Forest, XGBoost, and Logistic Regression models. The importance of environmental factors like wind speed and rainfall in risk prediction was highlighted by the Random Forest model's exceptional accuracy and recall. By offering a comprehensive and multifaceted approach, this research covers in the gaps in the current single-source models and opens the door for risk assessment techniques that are more accurate and comprehensible. Incorporating more telematics factors and investigating deep learning models for better forecasts are future goals of this research.**

I. Introduction

The way that driving behavior is monitored and how it is evaluated has been completely transformed by the rapid growth of telematics and the sensor technologies. Road conditions and weather have a significant impact on risk and while telematics data that covers variables like vehicle speed, acceleration, and location—offers useful data about the driving habits. Current risk prediction models that frequently overlook the significance of contextual factors like weather and the road characteristics in preference for depending exclusively on the telematics data and even though there is a wealth of data available. This lack of integration makes it difficult to evaluate the driving risks accurately and interpretably. Particularly in real-world situations when circumstances are complex and constantly shifting over the time.

In this project, we overcome these limitations by including the environmental elements and telematics data to create the effective machine learning models for driving risk prediction. Improving forecast accuracy through the integration of data from different sources and the use of advanced methods of modeling is the main objective of our research. We created a comprehensive dataset with over 200,000 observations by collecting meteorological data from the public sources, telemetry data from both car sensors and traffic accident records. To get the dataset ready for the machine learning and preprocessing techniques such as feature engineering, and imputation of missing values were carried out. Furthermore, a clear goal variable for the model training was made possible by the establishment of risk using the domain-specific thresholds for the environmental and telematics data.

The effectiveness of the the combined dataset was assessed using three machine learning models: XGBoost, Random Forest, and Logistic Regression. To ensure the equal representation of the risk classes and a stratified train-test split was used for both the training and validation. The models were optimized for measures that are including accuracy, recall, F1-score and AUC-ROC by extensive hyperparameter adjustment. The findings indicate that driving risk is greatly affected by environmental conditions such as rainfall and wind speed. In addition to highlighting the value of combining both environmental and telematics data and this research provides a scalable foundation for the future advances in risk modeling and such the addition of real-time data and deep learning.

II. Literature Review

A. Telematics and Risk Prediction

A new method for evaluating the driving behavior and predicting related dangers is telematics. Telematics data is crucial for tracking the vehicle statistics that including the speed, acceleration and the mileage that is according to several studies. Using an analysis of high-frequency driving actions and Li et al. (2023) showed how the important telematics is in forecasting the probability of an accident. Although these models give us an initial foundation for understanding the driver behavior and they frequently overlook contextual elements like road conditions and weather. Due to challenges with the gathering and integrating data, Bian et al. (2018) maintained that while adding the contextual factors that might greatly enhance of prediction performance and they are not used to their full potential.

B. Environmental Factors in Risk Assessment

Road conditions, rainfall, wind speed, and other environmental elements all have a significant impact on driving safety. The importance of incorporating the meteorological data into risk prediction models was highlighted by research conducted by Michelaraki et al. in 2021. However, evaluating and coordinating the real-time environmental data with the telematics inputs are frequently challenges for current methods. Further study has shown that bad weather such as heavy rainfall and

ice roads that can be significantly raise the probability of accidents. Although these results highlight the significance of these factors and there is still a lack of effective and complete integration of these variables into prediction models.

C. Machine Learning Models for Risk Prediction

In the field of risk assessment and machine learning models have been widely used for prediction tasks. The capacity of Random Forest and XGBoost to capture the intricate correlations between variables is well recognized. These models that performed better than conventional regression addresses when applied by Liu et al. (2017) to assess the telematics data. Although being simpler and the logistic regression provides interpretability and a strong baseline for the classification problems. Despite their effectiveness and the current research frequently ignores hyperparameter optimization and overfitting of the problems and which limits the models capacity to be widely used.

D. Gaps in Existing Work

While a lot of research has been done in environmental risk modeling and telematics data. There are still a lot of unanswered questions that are driving hazards are seen in fragments since the majority of research mostly depend on data from a single source either environmental or telematics. Class imbalance in datasets and which frequently moves predictions toward majority classes is also not given sufficient attention. Furthermore, very few studies include derived variables that might improve the model prediction ability that including speed on high-risk routes or acceleration during rain. The goal of this research is to close these gaps by utilizing the cutting-edge machine learning techniques and building an integrated dataset to produce accurate and the understandable risk predictions.

III. WORKFLOW

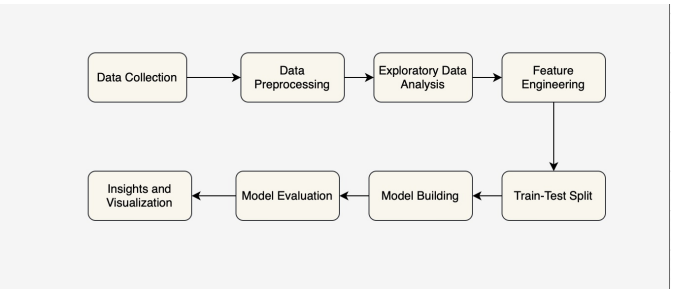

Fig. 1. Workflow

IV. OBJECTIVES OF THE STUDY

The primary objective of this project is to forecast driving and the roadway risks by combining environmental and telemetry data. The goal of the project is to create the machine learning models that can use factors like road features, weather and the vehicle speed to efficiently detect the high-risk situations. In order to increase the forecast accuracy and dependability, the project also prioritizes of tackling class imbalance concerns, limiting overfitting, and guaranteeing the model interpretability. The ultimate goal of the research is to support improved risk assessment methods and safer the driving habits.

V. DATA COLLECTION

Three main sources that were used to generate the information for this project: weather, telemetry, and traffic accident data. Vehicle data such as speed, acceleration and placement are included in the telematics data, which is taken from the Kaggle Automobile Telematics Dataset. Weather variables including temperature, precipitation, and wind speed are provided by the Weather LTSF Dataset. Accessible through Kaggle's Road Accident Data, road accident data provides information on accident sites and trends.

To ensure the compatibility for integration and eliminate missing values that each dataset received preprocessing. The final integrated dataset contains 213,792 rows and 21 features that are combining telematics and environmental data to build a robust foundation for predictive modeling.

VI. EXPLORATORY DATA ANALYSIS (EDA) AND HYPOTHESES FOR THE STUDY

A. Exploratory data analysis

The objective of the exploratory data analysis (EDA) was to learn more about the dataset and the correlations between the variables. Trends and irregularities in the data were found by analyzing the important statistics, distributions, and the correlations of the data. Significant patterns were revealed via data visualizations that including the graphs like heatmaps, scatter plots, and histograms. These included the effects of the rainfall, wind speed and the vehicle speed on danger levels. To address the class imbalance issues and the distribution of the target variable was carefully analyzed that guaranteeing the dataset was suitable for machine learning modeling. Also, In order to eliminate the multicollinearity and improve model performance then feature correlations were also examined.

B. Hypotheses for the Study

The following hypotheses serve as the foundation for the study: (1) Telematics variables such as vehicle acceleration and speed that have a big impact on driving danger. (2) Environmental elements like wind speed and rainfall are major contributors to an increase in road hazards. (3) Utilizing the environmental and telematics data combined yields a more accurate risk prediction than utilizing each source alone. (4) In predicting driving hazards and advanced machine learning methods like Random Forest and XGBoost can perform better than more conventional models like Logistic Regression while maintaining the interpretability and preventing overfitting.

C. Data Analytics

Data analytics was crucial to this study's results and predictive model development. To find the important trends

and connections that are affecting driving hazards for in-depth examination of the combined dataset of telematics and environmental factors was conducted. To make sure the data was ready for the model training and statistical techniques were used to investigate data distributions, correlations, and outliers as well.

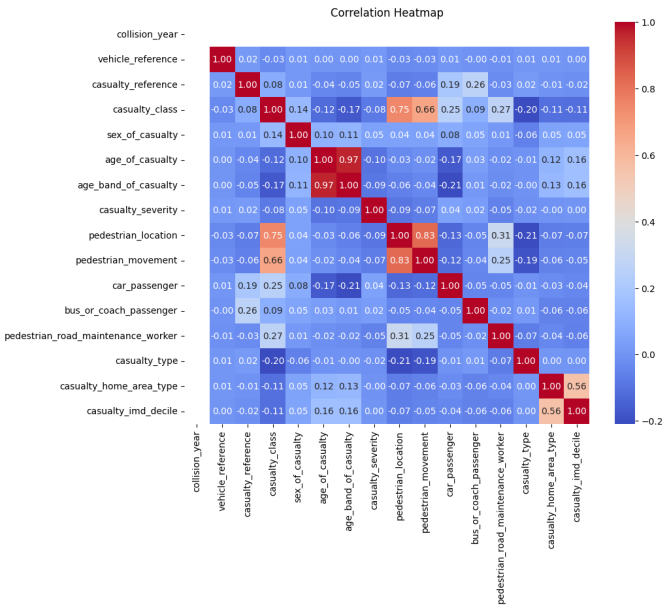

Fig. 2. Heatmap

Important analytic methods that included using graphs like heatmaps and histograms to identify the patterns and then computing descriptive statistics for describing variable features. Furthermore, derived variables like ”speed during rain” and ”acceleration on high-risk roads” were created via feature engineering that are improved the dataset’s prediction power.

D. Data Visualization and Results Report

The project evaluated the performance of the Random Forest, XGBoost, and Logistic Regression models for the risk prediction that using environmental and telemetry data. Metrics including Mean Absolute Error (MAE), F1-Score, AUC-ROC, Precision and Recall were used to assess each model. With an AUC-ROC of 0.75 and an MAE of 0.30 for logistic regression model provided a baseline that demonstrated a reasonable level of predictive accuracy. Random Forest and XGBoost showed better performance with AUC-ROC scores of 0.79 and 0.78 respectively, and lower MAE values of 0.25 and 0.24. These models ability to use feature importance and the capture non-linear correlations in the data allowed them to show their power in recognizing the high-risk scenarios.

The models relative efficacy is shown in the AUC-ROC graph. Despite the strong performance of all three models, Random Forest and XGBoost were able to find an acceptable balance between the precision and recall that guaranteeing good predicted accuracy for both the low-risk and high-risk categories. The findings highlight how crucial it is to use

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Logistic Regression Performance:
Precision: 0.65
Recall: 0.85
F1 Score: 0.74
AUC-ROC: 0.75
Mean Absolute Error (MAE): 0.30

Random Forest Performance:
Precision: 0.68
Recall: 0.97
F1 Score: 0.80
AUC-ROC: 0.79
Mean Absolute Error (MAE): 0.25

XGBoost Performance:
Precision: 0.68
Recall: 0.96
F1 Score: 0.80
AUC-ROC: 0.78
Mean Absolute Error (MAE): 0.24
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Fig. 3. Model Evaluation

sophisticated ensemble techniques for calculating the telematics risk. In addition to validating the selected models and this comprehensive evaluation opens the door for additional adjustment to enhance prediction performance.

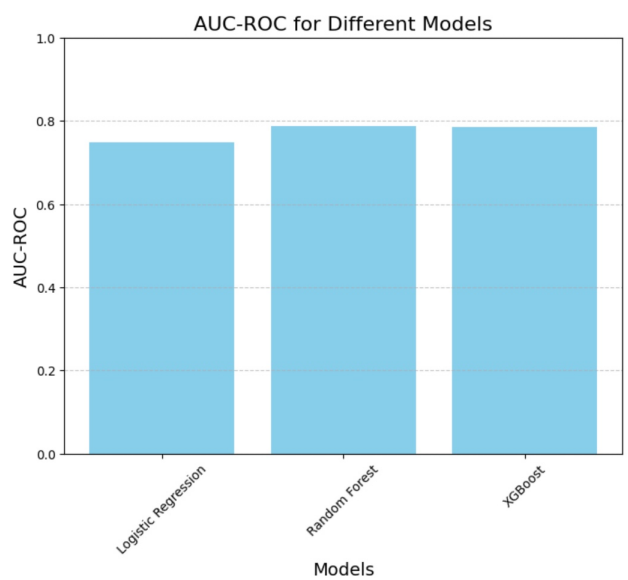

Fig. 4. Metrics

VII. Conclusion

In conclusion, this project effectively used the environmental and telematics data to use machine learning models to forecast the risk levels. The findings showed that in terms of predicted accuracy and dependability of the ensemble models

such as Random Forest and XGBoost performed better than Logistic Regression. It was emphasized how crucial environmental elements like rainfall and wind speed are to enhancing risk assessment. With potential applications in insurance and transportation safety, these findings enhance the field of telematics-based risk modeling. Future research that can be concentrate on investigating more sophisticated deep learning methods and that integrating more data sources.

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